

Arnav Vikram Shinde

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EDUCATION

Shri G.S. Institute of Technology And Science, Mechanical Eng.	06/2027 Indore, M.P.
Vidya Bhavan, 12th	03/2023 Ujjain, M.P.
Christu Jyotu Convent School, 10th	03/2021 Ujjain, M.P.

SKILLS

CAD (Creo, SolidWorks, Solidedge, NX CAD) | **CAE/FEA** (Ansys Mechanical/Workbench, Modal, Random Vibration, Thermal, Static Structural, Contact Mechanics, Rigid Body Dynamics) | **Multi Body Dynamics** (MSC Adams, IPG CarMaker, Ride Analysis, Fourpost Rig Simulation, Drop Test)

EXPERIENCE

Gabriel India Ltd. 🌐, Intern	05/2025 – 07/2025 Dewas, M.P.
- Acquired knowledge of suspension systems and damper functionality. - Studied supply chain management principles and manufacturing line operations & gained expertise in Design Verification Plan (DVP) and Design Failure Mode and Effects Analysis (DFMEA).	
Avinya MilTech 🌐, Intern	06/2026 – 07/2026 Indore
- Developed precision engineering drawings and 3D models of optical and mechanical components in Solid Edge, including a military-grade, waterproof enclosure for a Jetson Nano Orin AI camera system integrating LiDAR, IR, and HD imaging modules. - Performed CAE modal, random vibration, and thermal analysis on a satellite-related assembly. - Contributed to developing a complete kinematic simulator (Creo, MSC ADAMS) for a CMP lens-polishing machine—designing a quick-return-free four-bar oscillating arm, deriving disc profile geometry, calculating material removal rates using Preston's equation, developing a predictive re-flattening algorithm, and selecting bearings to optimize machine life—resulting in an app that accurately predicts machine behavior.	
Team GSRacers BAJA 🌐, Team Lead	03/2025 – 02/2026 Indore, M.P.
- Served as team's Technical Head, Vehicle Dynamics Head, Virtual Dynamics for 2026 e-Baja event. - Guided AIR1 to the Virtual Dynamic department.	

PROJECTS

Reaction Wheel Attitude Control System	02/2026 – 04/2026
- Designed and CAD-modeled a reaction-wheel ACS for a nanosatellite, deriving torque/momentum budgets and moment-of-inertia calculations from rigid-body attitude dynamics - Built an MBD simulation of the attitude control loop and validated simulated maneuver response against theoretical predictions - Performed Modal and Random Vibration Analysis in ANSYS to verify structural integrity and natural frequencies under launch loads	

AWARDS

AIR 1 in Virtual Dynamics (Validation Event) E-BAJA 2026, SAE BAJA	01/2026
AIR 7 overall in M-BAJA 2024, AIR 3 in Physical Dynamics in M-BAJA 202, SAE BAJA	01/2024